



Docker

Cheatsheet

*The secret to mastering Docker is
NOT what you think*



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Core Docker CLI



`docker build` creates an image from a Dockerfile.



code

```
1 docker build -t myapp:latest .
```



`docker run` launches a container from an image.



code

```
1 docker run -d -p 3000:3000 myapp
```



`docker ps` shows running containers, add **`-a`** to show all.



code

```
1 docker ps -a
```





Dockerfile Essentials



`FROM` sets the base image.



code

```
1 FROM node:18-alpine
```



`COPY` copies files into the container.



code

```
1 COPY . /app
```



`CMD` defines the default command.



code

```
1 CMD ["npm", "start"]
```





Docker Concepts



Image is a snapshot of your app including dependencies.



Container is a running instance of an image.



Volumes persist data even when containers are removed.



code

```
1 docker run -v /data:/app/data myapp
```

Use volumes to avoid losing your app's data after a container is stopped or removed.





Docker Compose Basics



``docker-compose up`` runs multi-container applications.



code

```
1 docker-compose up -d
```



Sample ``docker-compose.yml`` for a Node app with MongoDB:



code

```
1 services:
2   app:
3     build: .
4     ports:
5       - '3000:3000'
6     mongo:
7       image: mongo
8     volumes:
9       - mongo-data:/data/db
10    volumes:
11    mongo-data:
```





Docker Networking & Ports



Port Binding maps container ports to host machine.



```
1 docker run -p 8080:80 nginx
```



Bridge Network is the default Docker network.



Connect containers with `--network` option.



```
1 docker network create mynet
2 docker run --network=mynet --name=db mongo
```





Dockerfile Optimizations



Use multi-stage builds to reduce image size.

```
1 FROM node:18 AS builder
2 WORKDIR /app
3 COPY . .
4 RUN npm install && npm run build
5
6 FROM nginx:alpine
7 COPY --from=builder /app/dist
  /usr/share/nginx/html
```



Use `.dockerignore` to exclude unnecessary files.

```
1 node_modules
2 *.log
3 dist
```





Docker Debugging & Logs



`docker logs` shows container output.



code

```
1 docker logs <container-id>
```



`docker exec` lets you run commands inside containers.



code

```
1 docker exec -it <container-id> /bin/sh
```



`docker inspect` returns detailed container info.



code

```
1 docker inspect <container-id>
```





Image Hosting & CI/CD



Docker Hub is the public registry for images.



code

```
1 docker tag myapp username/myapp:1.0
2 docker push username/myapp:1.0
```



Use in CI/CD with GitHub Actions, GitLab CI, etc.



code

```
1 jobs:
2   docker:
3     runs-on: ubuntu-latest
4     steps:
5       - uses: actions/checkout@v2
6       - name: Build & Push
7         run: |
8           docker build -t myapp .
9           docker push username/myapp
```





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